thm.:

def.:

primitive notion: (undef.)

axiom: (postulates)
— facts ✓ without proof

statement: (proposition
— a sentence which is ✓ or ✗
but. not both

universal statement: (false) $\forall f(x)$ on $\mathbb{R}$ is cont.

(true) $\exists f(x)$ on $\mathbb{R}$ is not cont.

lemma $\subset$ thm.

corollary (上 切)

conjecture — T → thm.
— F

定 (Coro.)

E.g. $\forall x, y \in \mathbb{N}$ and x, y are odd,
$x+y$ is even

def. $m \in \mathbb{R}$ is an even int. $\Longleftrightarrow \exists k \in \mathbb{N}$ s.t. $m = 2k$

- - - odd - - - .. $m = 2k+1$

Proof:

$\forall m, n \in \mathbb{N}^*$

let m, n be odd int.

$\Rightarrow \exists k \in \mathbb{N}^*$, s.t. $m = 2k+1$

$\exists j$ - - -

$\Rightarrow m+n = 2(k+j+1)$

$k, j, 1 \in \mathbb{N}^*$

$\Rightarrow (k+j+1) \in \mathbb{N}^*$

$\Rightarrow$ by def., $m+n$ is even

false conjecture - e.g.:

$\forall n \in \mathbb{Z}$ and $n > 0$,

$n^2 + n + 41$ is prime → 质

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$P, Q, R, \ldots$ statements (prop.'s)

?

truth tabels

P	Q	(and) $P \wedge Q$	(or) $P \vee Q$	(not) $\neg P$	$\neg Q$	$P \Rightarrow Q$	$P \Leftarrow Q$	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$
✓	✓	✓	✓	✗	⋮	✓	✓	…	✗
✓	✗	✗	✓	✗		✗	✗		✓
✗	✓	✗	✓	✓		✓ Q	✗		✓
✗	✗	✗	✗	✓		✓	✓		✓

$(P \Leftrightarrow Q) \Leftrightarrow (P \Rightarrow Q) \wedge (P \Leftarrow Q)$

(T if) $(R \vee Q) \Leftrightarrow (\neg R \Rightarrow Q)$

$(P \Rightarrow Q) \Leftrightarrow (\neg \underline{P} \quad \overset{Q}{-})$

$\neg(P \Rightarrow Q) \Leftrightarrow \neg(\neg P \vee Q) \Leftrightarrow P \wedge \neg Q$

$\neg$ (if P then Q) $\Leftrightarrow P \wedge \neg Q$

converse of $P \Rightarrow Q$: $Q \Rightarrow P$

contrapositive of $P \Rightarrow Q$; $\neg Q \Rightarrow \neg P$

$(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$

de Morgan

$\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$

$\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$

direct proof of $(P \Rightarrow Q)$

1、Assume P is ✓

...

(prove Q is ✓)

E.g. The product of 2 odd int. is odd

$\Leftrightarrow$ $\forall$ odd m, n, $(m \cdot n)$ is odd

Proof: Let $\forall$ odd m, n

def. $\Rightarrow$ $\exists k \in \mathbb{Z}$ s.t. $m = 2k+1$,

$\exists j$ . . . $n = 2j+1$

$m+n = (\sim) + (\sim) = 2(\sim)$

def. $\Rightarrow$ $(m+n)$ is odd

Corollary: $\forall$ odd n, n^2 is odd

($\Leftrightarrow \forall$ even n, n^2 is even)

(contrapositive)

Proof: (见 [illegible])

$$n = 2k+1$$

$$n^2 = (4k^2+4k)+1$$

$$= 2(2k^2+2k)+1$$

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$P \Rightarrow Q$

证明方法：1、直接：assume P, prove Q

2、contrapositive $(\Leftrightarrow)$ $\neg Q \Rightarrow \neg P$
反证
(Assume Q is false $(\neg Q)$,
prove P is false $(\neg P)$)

3、by contradiction: Prove stmt S is true:
反例
Assume $\neg S$
↓ arrive at a contradiction
↓
S is true

S : $P \Rightarrow Q$
$P \wedge \neg Q$ $\Rightarrow$ $\neg Q$ is true
→用反证→ $\neg P$
→ $P \wedge \neg P$ 矛盾 → $(P \Rightarrow Q)$ is true

E.g. $\forall x \in \mathbb{Z}$, $\underbrace{3x+8 \text{ is even}}_{}^{\neg P}$ $\Rightarrow$ x is $\overset{\neg Q}{\text{even}}$ → S

用反证：$\Updownarrow$ $\forall x \in \mathbb{Z}$, x is odd $\Rightarrow$ $3x+8$ is odd

... $\underbrace{\exists k \in \mathbb{N}^*}_{\text{is int}}$, s.t. $x = 2k+1$

--- $\Downarrow$ is int

$3x+8 = $ ---- $\Rightarrow$ is odd

$\Rightarrow$ ~~

用反例：Assume that S is false
→ the negation is true
否
→ $\exists x \in \mathbb{Z}$, s.t. $3x+8$ is even $\Rightarrow$ x is odd
→ ... → $3x+8$ is odd → contra~ → S is true

直接证：...

E.g.: $\forall x, y \in \mathbb{Z}$, xy is even $\Longleftrightarrow$ (x is even $\vee$ y is even)

1、(証 $\Leftarrow$)

Assume that x is even $\vee$ y is even

WLOG, we may assume that x is even

(∵ if we exchange x and y,
same process,
so we have symmetry)

(x is even)
(by def.) $\Longrightarrow$ $\exists k \in \mathbb{N}$, s.t. $x = 2k$

$\Rightarrow xy = (2k)y = 2(ky)$ ($\mathbb{N}$, $\in \mathbb{N}$) $\Rightarrow ky \in \mathbb{N}$

$\Rightarrow ky$ is even

2、(証 $\Rightarrow$)

(contrapositive)

($\forall x, y \in \mathbb{Z}$, ($x$ is odd $\wedge$ y is odd) $\nearrow$ ($\neg Q$)

$\Rightarrow xy$ is odd $\searrow$ ($\neg P$)

$\Updownarrow$

($\forall x, y \in \mathbb{Z}$, xy is even $\Rightarrow$ (x is even $\vee$ y is even)

$\searrow$ (P) $\searrow$ (Q)

Rational number

$x \in \mathbb{R}$

$\underbrace{x \text{ is rational}}_{\substack{\downarrow \\ (x \in \mathbb{Q})}} \Longleftrightarrow \exists m, k \in \mathbb{Z},\ k \neq 0,\ \text{s.t. } x = \frac{m}{k}$

$(\Longleftrightarrow \exists m, k \in \mathbb{Z},\ k > 0,\ \text{s.t. } x = \sim)$

累证: $\underbrace{\forall n \in \mathbb{R},\ \forall x \in \mathbb{R},\ x^n \notin \mathbb{Q} \Rightarrow x \notin \mathbb{Q}}$

$(\Longleftrightarrow \forall n \sim,\ \forall x \sim,\ x \in \mathbb{Q} \Rightarrow x^n \in \mathbb{Q})$

① (contrapo.)

证: Let $\underline{n \in \mathbb{N}}$ and $\underline{x \in \mathbb{R}}$

Assume that $\underline{x \in \mathbb{Q}}$

$\Rightarrow \exists k \in \mathbb{N},\ m \in \mathbb{Z},\ \text{s.t. } x = \frac{m}{k}$

$\Rightarrow x^n = \frac{m^n}{k^n} \begin{array}{l} \to \in \mathbb{Z} \\ \to \in \mathbb{N} \end{array} \Rightarrow x^n \in \mathbb{Q}$

② (contradiction)

negation: $\exists n \sim,\ \exists x \sim,\ \text{s.t. } x^n \notin \mathbb{Q}$

([illegible]略)

③ (proof by cases)

([illegible]略)